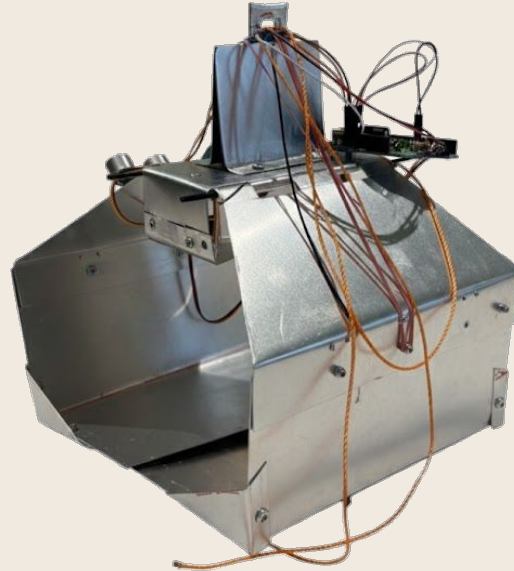


SCHKËUWP

A helping hand that never tires



Team A6

Ming Gao, Kyrin Chen, Marcus Kwong,
Adrian Lee, Charles Xiong, Si Wu

OBJECTIVE & DESIGN STRATEGY

Objective

To develop a versatile mechanical claw/retrieval system that boasts robust grasping capabilities with operational efficiency to handle and retrieve a wide array of objects with quick cycle times.

Strategy

Our team chose to design a retrieval system that could both retrieve a great number of symmetrical items, or excel at picking up irregular shaped items, allowing us to maximize points in rounds 1 (variety) and 2 (bulk round), since our team had the most control over these rounds.

Target Design Specifications

Requirements

Can automatically open and close to grab objects

Can be constructed with at most 6 pieces of 8x8-inch sheet metal

Overall structure is durable for repetitive uses

Arduino and servo components can be integrated

Evaluation Criteria

Retrieves numerous items in one cycle

Is not prone to misalignment after use

Objects will not fall or roll out of the retrieval system

Has a large receptacle volume

DESIGN PROCESS

02 EARLY PROTOTYPING

Building simple and low-cost models to quickly visualize form and test basic mechanical functions



01 CONCEPT GENERATION

Creating a pool of diverse ideas through individual and collaborative sketching

ITERATION

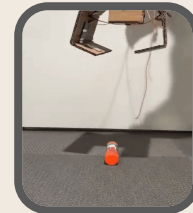
Utilizing new information to revisit and refine concepts through entire process

03 EVALUATING IDEAS

Screening, ranking, and scoring concepts to narrow down most viable options

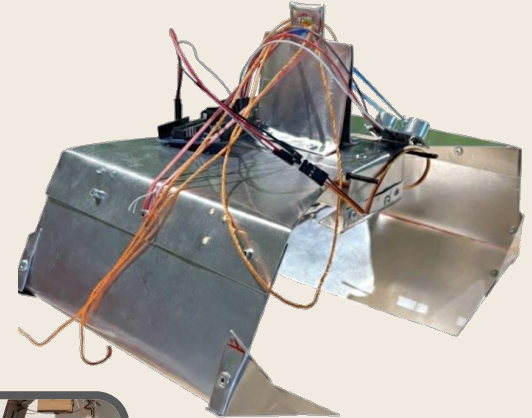
04 LATE PROTOTYPING

Developing comprehensive, high-cost prototypes for final two competing concepts



05 FINAL DEVELOPMENT

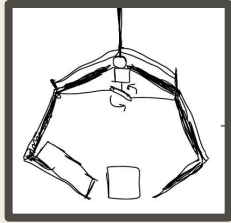
Developing optimal design and refining it for final production and use



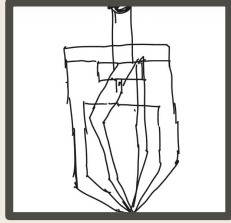
POOL OF CONCEPTS

Each member proposed two concepts, for a total of 12. Each member then chose one concept to prototype

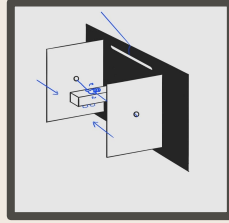
Advanced



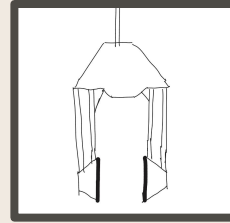
Idea 1



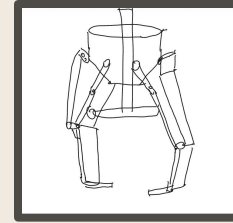
Idea 2



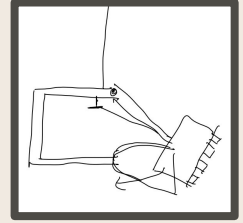
Idea 3



Idea 4

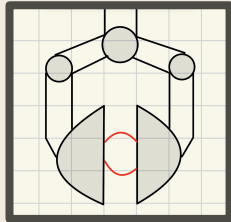


Idea 5

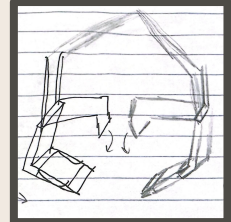


Idea 6

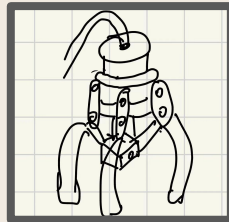
Not Advanced



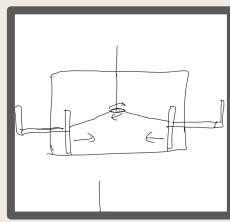
Idea 1



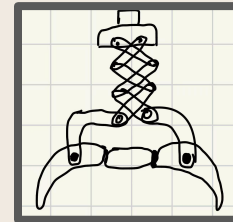
Idea 2



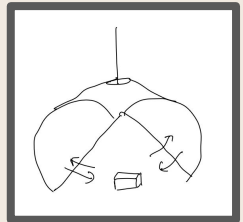
Idea 3



Idea 4



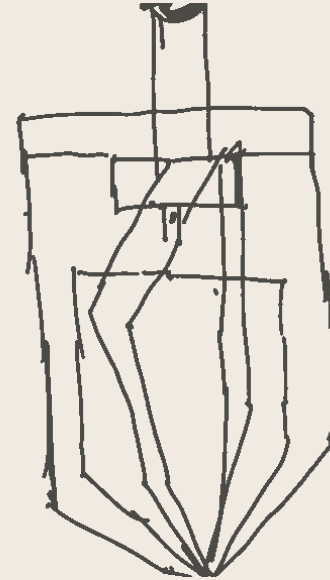
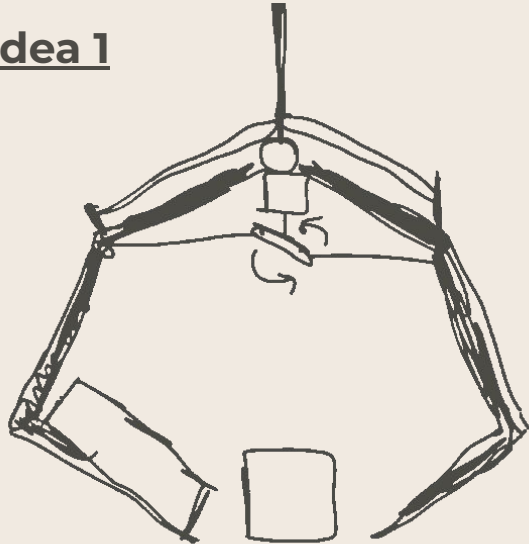
Idea 5



Idea 6

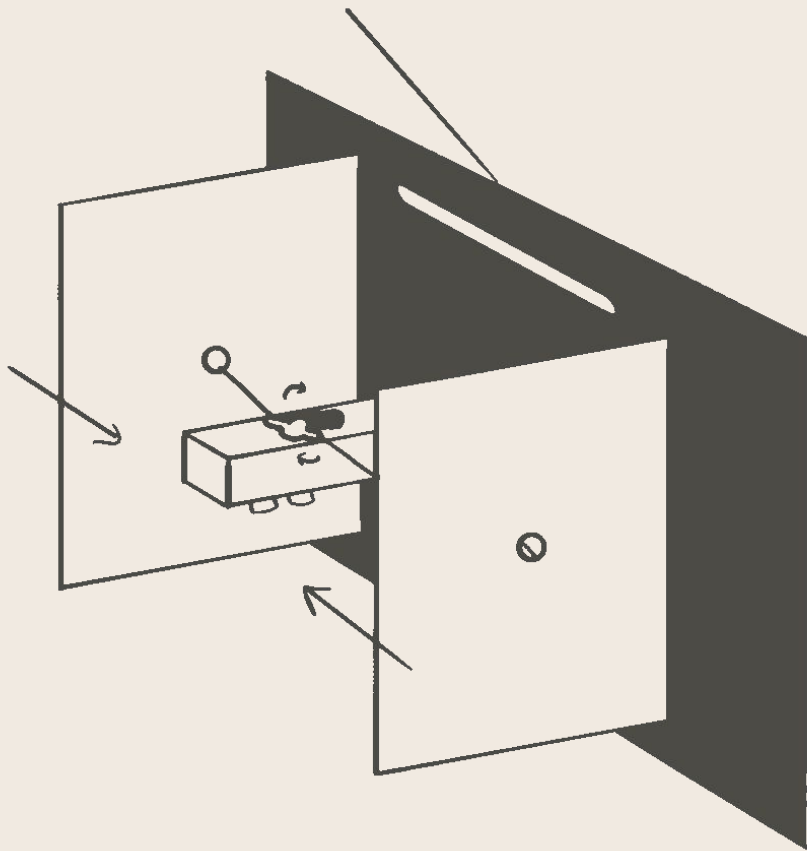
Idea 1 uses a **symmetrical multi-link design**. It utilizes a **central rotating joint** that converts a servo motor motion into mechanical arm movement. The claw heads are **large plates** that fold into a receptacle, perfect for **“scooping”** up irregular and smaller objects.

Idea 1



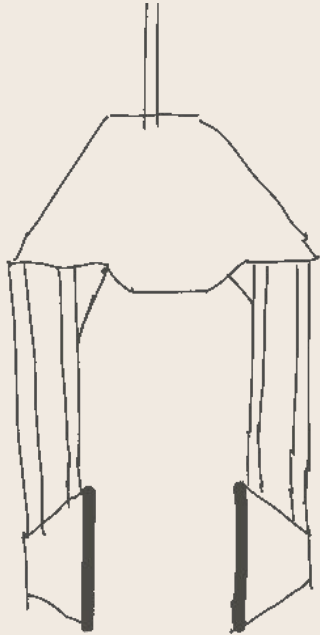
Idea 2

Idea 2 resembles a more **conventional** claw design, focused around **angled, inward-folding** arms that **converge** toward a central point when the mechanism is activated. Idea 2 is especially effective at retrieving irregular or tapered shaped objects.



Idea 3

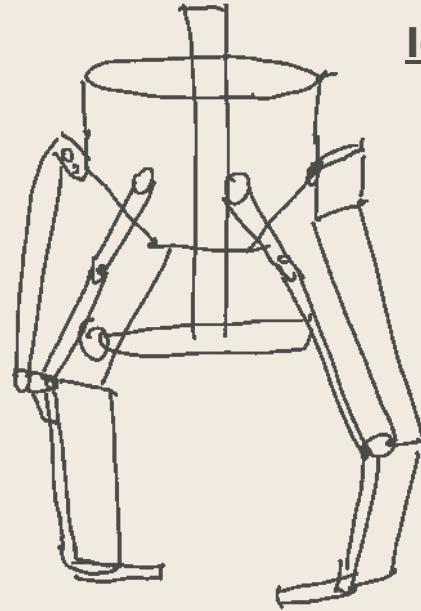
Idea 3 is our most **unconventional** design, centered around a **sliding linkage** mechanism. It utilizes a central block which **houses a servo motor** that closes **two large plates**. The motion is guided by slits on the body, allowing smooth and **controlled clamping** between the two plates. This design would be effective for objects with large smooth faces, but struggle against small and irregular items.



Idea 4

Idea 4 makes use of two **vertical arms** driven from the top. **Flat gripping pads** are attached at the lower end of each arm, great for picking up objects with flat edges or slippery sides.

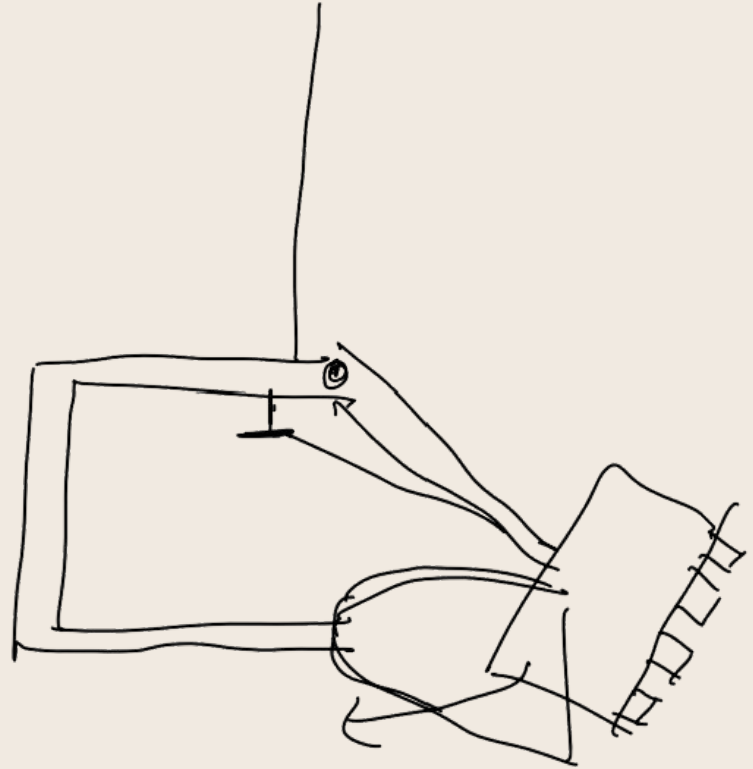
Idea 5 is another typical claw design, using a **linkage** driven claw that is **actuated from the top**. The arms **swing inward to grasp** an object, providing precise control.



Idea 5

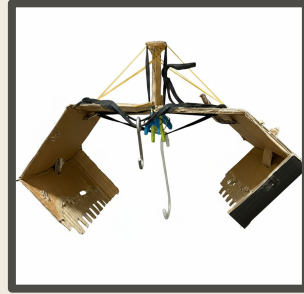
Idea 6

Idea 6 works like a broom: **one wide plate** slide underneath the object, then the **top fork** “sweeps” it onto the plates and into the center. Once aligned, the forks apply gentle pressure to hold it in place and prevent it from rolling or slipping out the ends. This is best for **soft or round objects**, like balloons or golf balls.



PHYSICAL PROTOTYPING

- Each member chose one of the advanced concepts to construct a physical prototype
- Physical prototypes allowed us better understand the form and function of each concept



Idea 1



Idea 2



Idea 3



Idea 4



Idea 5



Idea 6

SCREENING

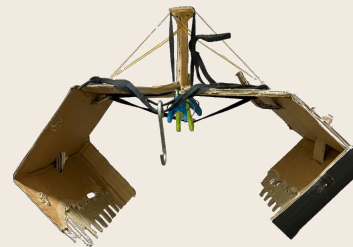
Green ideas are advanced

Requirement	Justification	Idea 1	Idea 2	Idea 3	Idea 4	Idea 5	Idea 6
Can automatically open and close to grab objects	The claw should perform an automatic retrieval using arduino and servo components	✓	✓	✓	✓	✓	✓
Can be constructed with at most 6 pieces of 8x8-in sheet metal	6 pieces is the maximum number of sheet metal available	✓	✓	✗	✗	✓	✓
Overall structure is durable for repetitive uses	The design and structure of the final product should not break too easily	✓	✓	✓	✗	✓	✗
Arduino and servo components can be integrated	The design allows for the use of an arduino and servo motor for automation	✓	✓	✗	✗	✓	✗

RANKING

Individual voting / 2 votes per person

Person	Idea 1	Idea 2	Idea 5
Ming G	✓		✓
Kyrin C	✓		✓
Marcus K	✓	✓	
Charles X		✓	✓
Adrian L	✓✓		
Si W	✓		✓
Total	6	2	4



Idea 1



Idea 5



*Comprehensive prototype of Idea 1 retrieving
a glue stick*

DEVELOPED PROTOTYPES

To more accurately score our final two concepts, we developed comprehensive prototypes for each.

This allowed us to conduct a side-by-side performance analysis against our evaluation criteria, ensuring our final selection was based on real data rather than intuition.

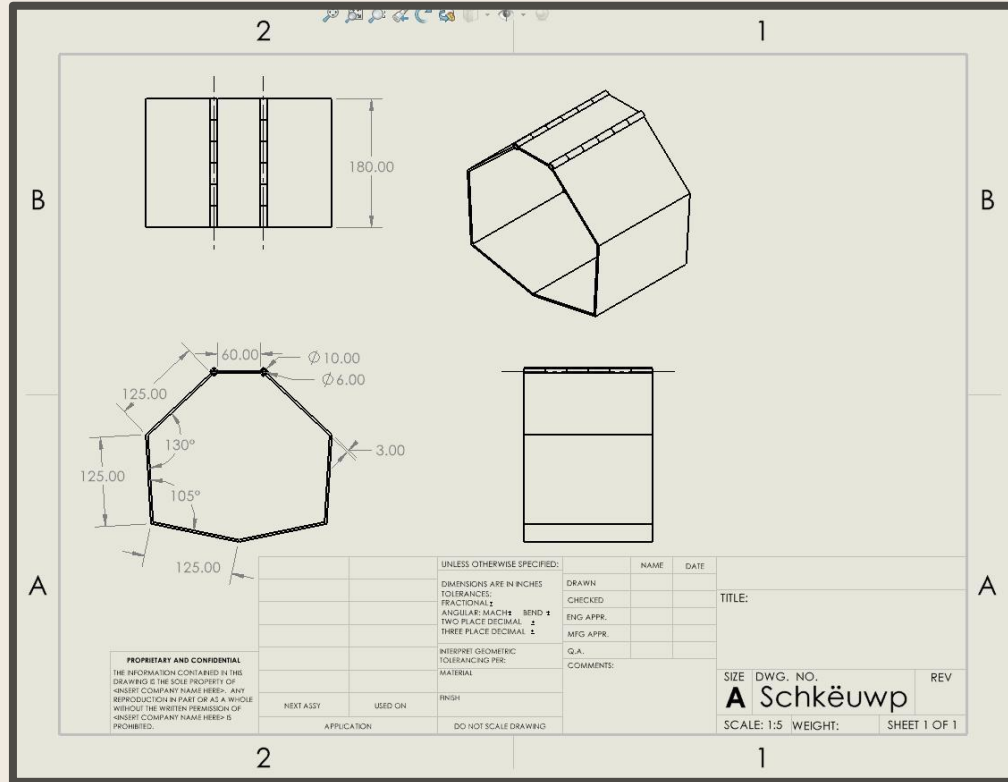
SCORING

Via weighted decision matrix

Evaluation Criteria	Justification	Weight	Idea 1		Idea 5	
			Raw Score	Scaled Score	Raw Score	Scaled Score
Retrieves numerous items in one cycle	Allows us to maximize points in the bulk round	20%	8	1.6	3	0.6
Is not prone to misalignment after use	Further retrieval can not be achieved with a misaligned claw	20%	8	2	5	1.25
Objects will not fall or roll out of the retrieval system	Decreases the amount of wasted time	25%	8	2	5	1.25
Has a large receptacle volume	Provides more flexibility on different shape of objects	35%	9	3.15	8	2.4
		Total		8.45		5.5

Weights for scoring were determined by team discussion informed by prototype testing and project goals.

ENGINEERING DRAWINGS



RISK ASSESSMENT & MANAGEMENT

Assessment

Risks were first identified individually and added to a classification table. Further risks were then identified through team discussion.

Risks were assessed using a risk matrix, ranking each according to its severity of impact and likelihood of occurrence. The product of these two scores provided a quantitative risk level.

Management

Any risk with a total score exceeding 9 was classified as high-priority, requiring monitoring by one or more team members.

RISK REGISTRAR

	External	Preventable	Strategic
Safety	Flying debris from another team leads to an eye injury	- Cuts from sharp sheet metal edges - Finger pinching from using tools	
Technical	Unexpected hardware failures during the competition	- Insufficient gripping force to hold heavy objects - Cardboard joints weaken with repeated use - Sheet metal and cardboard interfaces tear due to poor construction - The claw head misaligns after the retrieval of a large or obscure object - Strings and rubber bands stretch with use, decreasing grip strength	- Reinforcing certain parts, making it stronger, but heavier - Using a wide and flat claw head improves the variety round performance, while sacrificing performance in the bulk round
Project Management	- Teammate absence due to an illness or injury - Limited access to tools or workshop time	- Missing internal deadlines due to poor communication and delegation - Manufacturing challenges delay the project's completion	Allocating more time to concept generation to improve concept quality, at the expense of reduced testing and iteration time
Operational	Competition is rescheduled, changing available time	- Batteries not charged before the competition - Software and hardware not tested and calibrated before the competition	Each of the team members knows how to operate the retrieval system designed

RISK MATRIX

Risks with asterisks (*) are monitored

Almost Certain	5	Strings and rubber bands stretch with use, decreasing grip strength	*Reinforcing metal layers of certain parts of the claw, making it stronger, but heavier	*Using a wide and flat claw head improves the variety round performance, while sacrificing performance in the bulk round		
Likely	4		Limited access to tools or workshop time	*Insufficient gripping force to hold heavy objects	*Unexpected hardware failures during the competition	
Possible	3		Cuts from sharp sheet metal edges Finger pinching from using tools	*Teammate absence due to an illness or injury	*The claw head misaligns after the retrieval of a large or obscure object	
Unlikely	2			Sheet metal and cardboard interfaces tear due to poor construction	Flying debris from another team leads to an eye injury	*Batteries not charged before the competition
Rare	1		Competition schedule changes, increasing or decreasing the time available		Missing internal deadlines due to poor communication and delegation	Claw spontaneously combusts
		1	2	3	4	5
		Negligible	Minor	Moderate	Major	Catastrophic

RISK MITIGATION STRATEGIES

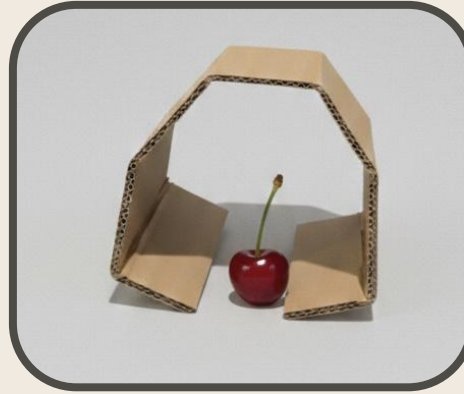
Person	Risk	Mitigation Strategy
Ming G	Batteries not charged before the competition	Check charge 6 hours before competition and carry fully charged backup battery packs
Kyrin C	Unexpected hardware failures during the competition	Conduct a stress test on all hardware components 6 hours before competition and carry backup hardware kit
Marcus K	The claw head misaligns after the retrieval of a large or obscure object	Manually align claw heads to default position after every retrieval cycle
Charles X	Reinforcing metal layers of certain parts of the claw, making it stronger, but heavier	Test different loads, recording where strength increases without exceeding the motor's torque
Adrian L	Using a wide and flat claw head improves the variety round performance, while sacrificing performance in the bulk round	Conduct timed trials with various objects with different sized claw heads
Si W	Insufficient gripping force to hold heavy objects	Test various high friction surfaces for implementation for claw

Optimal Claw Head Size

Having decided on a design that utilized two scooping heads instead of a more traditional claw, we needed to decide how wide to make our claw heads to optimally pick up the specified objects in the competition.

To determine the ideal dimensions, we prototyped several claw heads with varying widths and conducted trials with competition conditions.

The data from our trials allowed us to identify 8 inches as being the width that offered the optimal balance between load volume, maneuverability, and alignment ease.



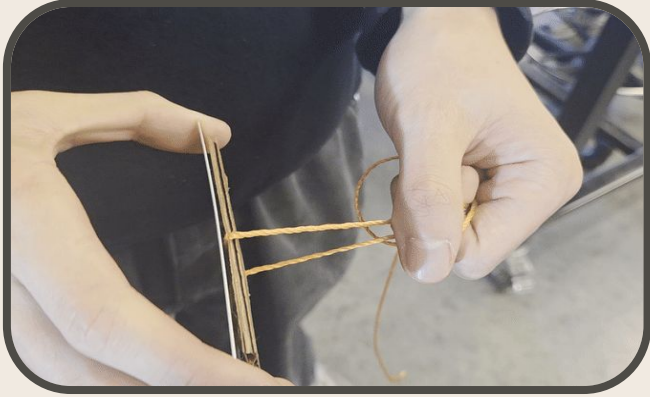
Claw head of 6 inches



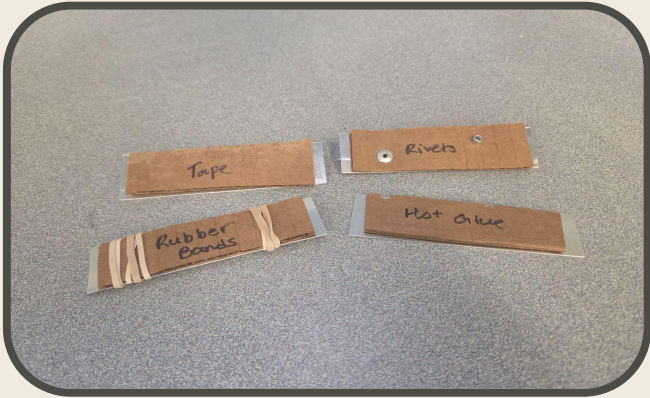
Claw Head of 8 inches



Number of objects retrieved for each claw head size



Testing strength of hot glued interface



Cardboard attached to sheet metal via different methods

Material Joint Test

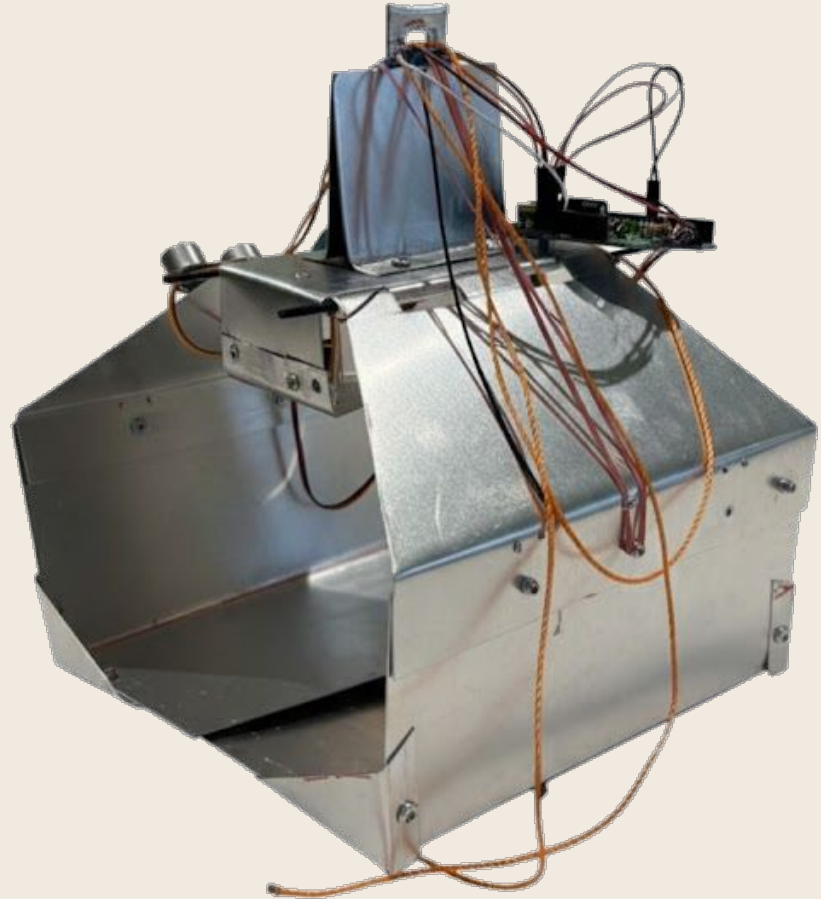
To protect our electronics from the impact of the claw striking the ground, we chose to use cardboard to house the electronic components and guide internal mechanisms. To find the best way to join these two materials, we qualitatively ranked several methods using a rope-pull test to see which bond resisted tearing between two 3x1-inch pieces of the two materials. While adhesives were easy to apply and kept the look of the claw clean, mechanical fasteners proved to be much stronger, leading us to use rivets in our final product to join cardboard and sheet metal together.

Rivets	Rubber Bands	Hot Glue	Tape
Exceptional	Adequate	Adequate	Negligible
Required much greater force than the other options	Contacts were held strong, but cardboard could slip	Cardboard came off once a certain force was reached	Similar to hot glue, but force required was less

Chosen Concept

Schkëuwp

The Schkëuwp (/skɛp/), is a **high-capacity, all-metal** retrieval system engineered for **bulk** object retrieval. Featuring **L-shaped claw heads**, the Schkëuwp excels at **securing irregularly shaped items** regardless of their orientation. With a generous receptacle volume of **8.5L**, it effortlessly allows for **large or unconventional** loads.



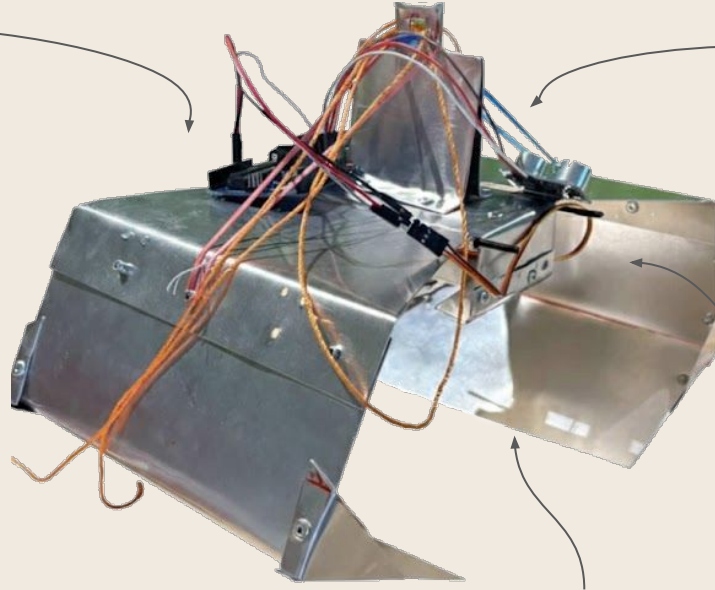
FINAL DESIGN

01 Arduino Automation

Integration of arduino system with sonar sensor and servo motor allows to close the claw with sufficient force

02 Receptacle Guards

Receptacle guards on the edges of the wide claw heads protect items from falling out during retrieval



05 Secured Rubber bands

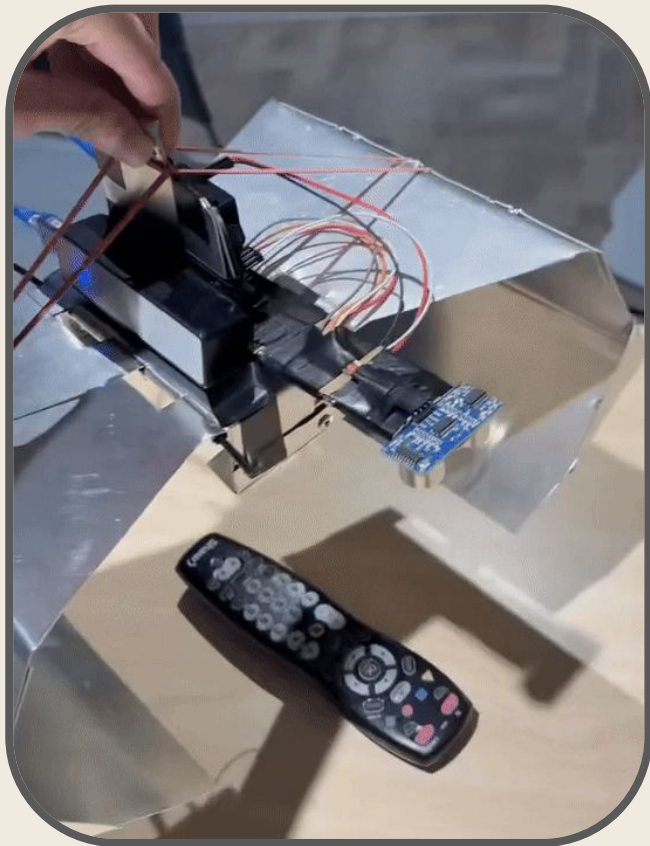
The two attached rubber bands on both sides allows to apply opposite force to servo motor. It builds up an automatic open and close system.

03 Smooth Hinges

Smooth hinges allow for repeated use without manual resetting

04 Wide L-Shaped Claw Heads

Wide claw heads allow the retrieval of multiple objects within one cycle



Retrieval of a TV remote with the Schkëuwp

COMPETITION PREPARATION

Team Practice

Our team extensively practiced retrieving various objects with the claw. Testing was done with one member holding the claw and moving it up and down.

Preliminary Checks

6 hours before the competition, each member conducted specific checks to ensure components were prepared.

- Battery pack checks
- Component checks
- Spare electronic kits check
- Mechanical components check

COMPETITION RESULTS

As predicted from our prototyping and technical investigations, the large claw heads successfully retrieved bulk and irregular loads with the flexibility and ease we expected. The claw also demonstrated high structural durability, with our electronics housings staying fixed thanks to the strong attachment method we chose with our risk analysis.

Despite the success of the physical design of our claw, our scores were limited due to operational challenges and a lack of familiarity with the provided crane's handling, which hindered our cycle efficiency. We also encountered a mechanical fault (see slide 26), which also hindered our performance.

Variety Round	Bulk Round	Cooperative Round
9 Objects	23 Foam Blocks	XYZ?



Retrieval of objects in round 1 and 2 of competition

COMPETITION REFLECTION

Competition Equipment

During the competition, it became evident that tests and practice runs should have used equipment from the competition. Our team had issues with adjusting the height and position of our claw with the provided crane. Our team only practiced using our arms and string, which was not similar to using the crane.

Improving Coordination

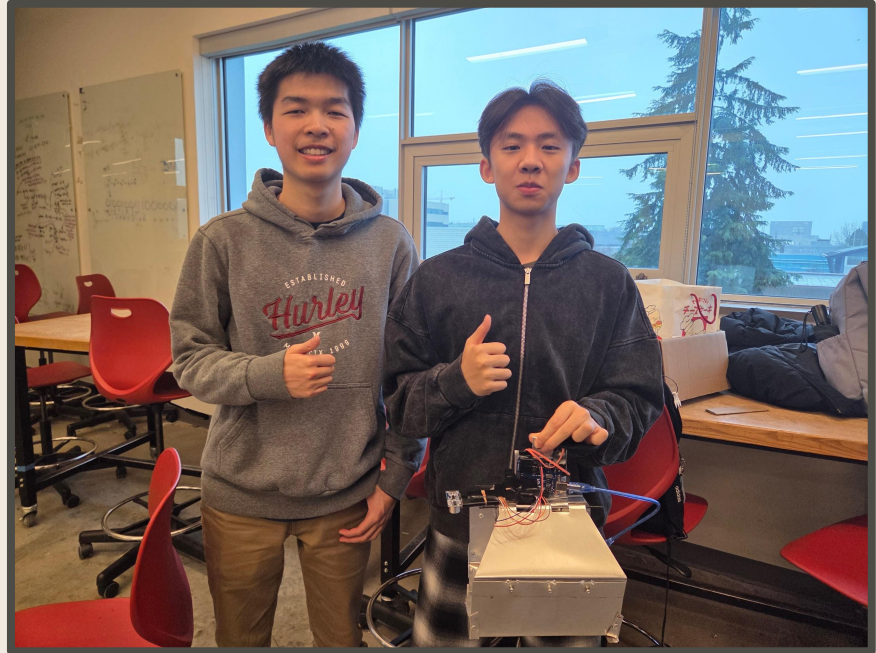
Given the challenges with positioning our claw, the level of coordination did not reflect what was seen during our practice. Specific team members became stressed and impatient during the competition rounds, leading to misaligned retrievals and improper positioning. Tests simulating more closely to the competition rounds may have made team members more mindful of the environment they would find themselves in.

Mechanical Fault

A mechanical mishap also occurred when a part of the floss wound up around the servo screw, possibly during the attachment of rubber bands. This reduced the jaw size of the claw for the first two rounds of the competition by around 30%, which greatly affected the claw's efficiency in both rounds.

MODULE REFLECTION

Module 5 served as a introduction to the multifaceted nature of engineering projects. It challenged our team to bridge different concepts, such as programming with Arduinos, fabricating in the workshop, the standards of engineering drawings, and creation of technical communication such as this e-poster. However, the most important thing our team witnessed was how our team dynamic evolved from a group of individuals to a performing collective unit, highlighting that the synergy of a team is just as important as technical skills. Our ability to maximize the use of a diverse skill set was only possible once our team established mutual trust and strong communication.



Post-competition photo with the Schkëuwp

Team Contribution

All team members contributed to the development of the e-poster. While everyone provided ideas, feedback, and content throughout the process, Marcus Kwong, Charles Xiong, and Si Wu were primarily responsible for assembling and refining the final e-poster. After the draft was completed, the full team reviewed it together and suggested edits to ensure clarity, consistency, and overall quality.

Ming Gao:

Kyrin Chen:

Marcus Kwong:

Adrian Lee:

Charles Xiong:

Si Wu:

GenAI ACKNOWLEDGEMENT

Generative AI was used throughout the production of this presentation as a tool to assist and refine in the following ways:

- To **remove backgrounds** and **upscale images**
- To **proofread** for **grammar** and **conventions**